Surgical Robotics & Navigation · Research Prototype

Oral and Maxillofacial Surgical Navigation (SMCNavi → OprNavi → OmfsNavi)

Integrated image-to-patient registration, surgical instrument tracking, and calibration in SMCNavi, then carried the result into XR surgical visualization on HoloLens.

Period 2023.07 - present | Evidence state Prototype · Ongoing | Portfolio tier Surgical Robotics & Navigation

The SMCNavi platform and HoloLens extension developed by DIGITRACK with Samsung Medical Center in a research collaboration. In 2026 it continued through OprNavi v0.1.0 into OmfsNavi, a C++ rebuild.



Figure 1. Full-length research-prototype demonstration connecting the HoloLens 2 digital twin with tracked instruments and image presentation.

Problem and system boundary

Oral and maxillofacial navigation must align medical images, patient and instrument coordinates, procedure-specific tools, and spatial presentation in one flow; fragmented functions make registration state and data flow difficult to inspect.

Owned decisions and implementation

My role

3D Medical Imaging · Surgical Navigation Developer

Designed the overall software architecture and served as the primary implementer for DICOM and 3D-model loading, MPR and 3D visualisation, the optical-tracker SDK and data pipeline, image/patient/marker/instrument transforms, patient registration and feedback, marker/non-standard/long-instrument calibration, six workflows and mirroring, HoloLens-PC communication and interaction, and phantom integration tests and verification tooling. I was effectively the sole developer, and implemented mirror-plane registration in August-September 2025. In 2026 I continued with a shared tracking service stabilisation, migration to tracker SDK v7.1.2, and the design and instrumentation of a navigation-latency measurement harness, built a hands-free control PoC (foot switch and voice, excluding risky commands such as registration and Collect), and rebuilt the work as OprNavi v0.1.0, a general case-based navigation (signed-distance safety margin, tip-to-surface guidance, per-point residual RMSE, registration presets, tracking record and replay, tracker SDK selection and diagnostics, checksum-manifest installer, 58 service tests), and OmfsNavi (Slicer 5.12.4, Qt 6, C++ loadable modules, module-independent tracking with per-feature readiness, an independent verification point not used for registration, a plan.json v1 surgical-plan importer, CTest, a mock backend). Compatibility of the plan.json importer with the digital occlusion workflow app (pre-operative planning) has not been verified.

SMCNavi platform overview

SMCNavi integrates DICOM and 3D-model loading, MPR and 3D visualisation, optical tracking, patient registration, instrument calibration, and procedure-specific UI in one custom 3D Slicer desktop platform. The HoloLens work is a separate PC-side extension connected to SMCNavi. Participation in earlier navigation work dates to 2023, while the main SMCNavi repository implementation is recorded from 2025. I worked with researchers on procedure-specific requirements and carried them into modules, installation packages, and user guidance.

Six oral and maxillofacial workflows

The following are implemented and demonstrated software workflows; they are not evidence of clinical efficacy.

- Maxillary tumour-removal navigation
- Mandibular tumour-removal navigation
- Bimaxillary-surgery navigation
- Mandibular-motion tracking
- Bone-graft placement
- Zygomatic-orbital fracture mirroring

Tracking, registration, and calibration

Built explicit transform paths between the patient, instruments, and image models so the inputs and feedback at each stage could be inspected.

- Optical-tracker SDK integration and the live tool/marker data pipeline
- Transform chain across image, patient, marker, and instrument frames
- Patient registration and registration-state feedback
- Patient and instrument marker calibration
- Non-standard and long-instrument calibration
- Mirror-plane registration with pivot-point setting
- 2026 stabilisation: a shared tracking service used across screens, and migration to tracker SDK v7.1.2
- 2026 stabilisation: design and instrumentation of a stage-by-stage navigation-latency measurement harness (no latency results claimed)

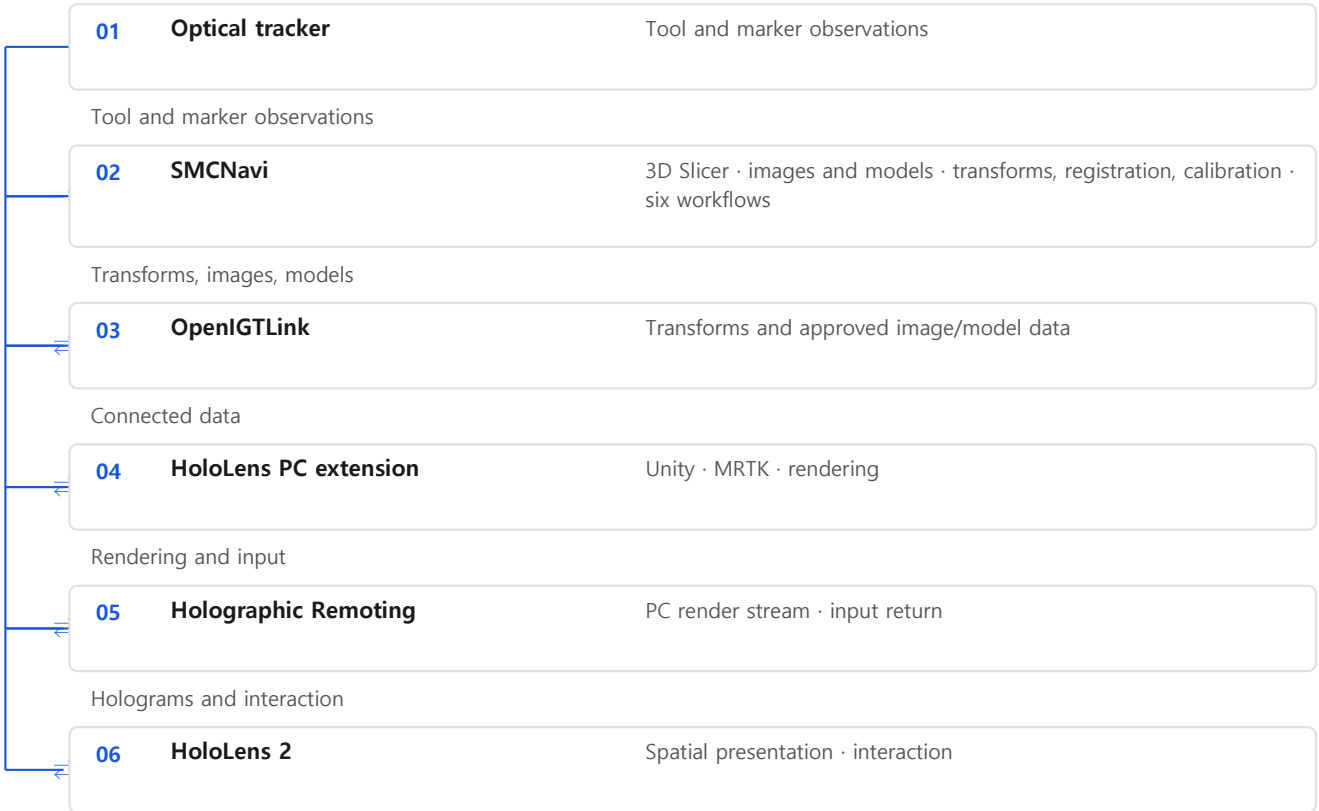
HoloLens spatial interface (2024-2025, ended)

Rendered SMCNavi-owned registration and workflow state in a separate PC extension and connected it to HoloLens 2 spatial presentation and interaction. The HoloLens path ran in 2024-2025 and has ended.

- Transform and approved image/model exchange through OpenIGTLink
- PC rendering and spatial placement with Unity and MRTK
- HoloLens 2 presentation through Holographic Remoting
- Return of HoloLens interaction input to the PC extension
- Phantom integration tests across optical tracking, SMCNavi, and HoloLens
- Nov 2025 - Jan 2026: Galaxy XR WebRTC mirroring experiment (screen capture → H.264 hardware encoding → WebRTC → Unity; no latency figures claimed)

From tracking observations to HoloLens interaction

Explanatory system-relationship diagram; it is not photographic, experimental, or clinical-outcome evidence.



SMCNavi-HoloLens path · Research prototype

Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

Two full-length videos and interface, coordinate-frame, instrument, and phantom figures show a working research prototype carrying position, model, image, and interaction data from SMCNavi through the HoloLens path.

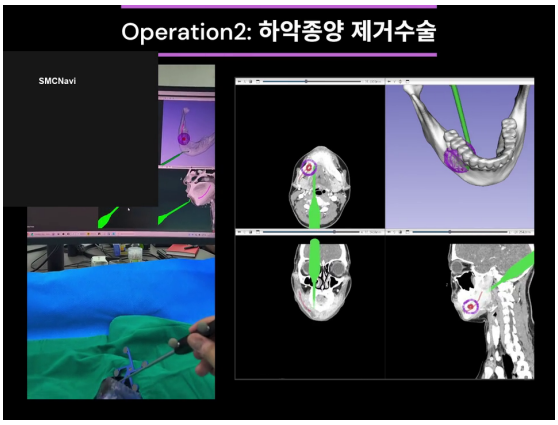


Figure 2. Full feature video moving through six oral and maxillofacial workflows in SMCNavi.



Figure 3. Integrated SMCNavi UI and HoloLens-PC connection view.



Figure 4. Six oral and maxillofacial software workflows.

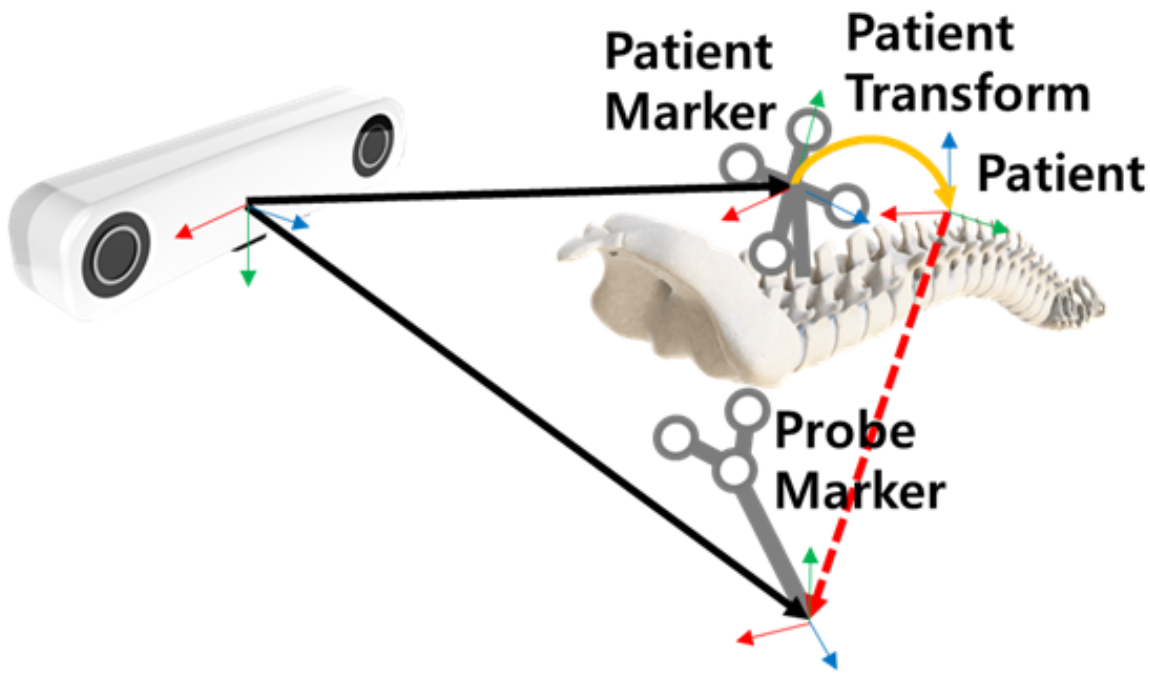


Figure 5. Coordinate frames: patient-marker and probe-marker transforms relative to the optical tracker.

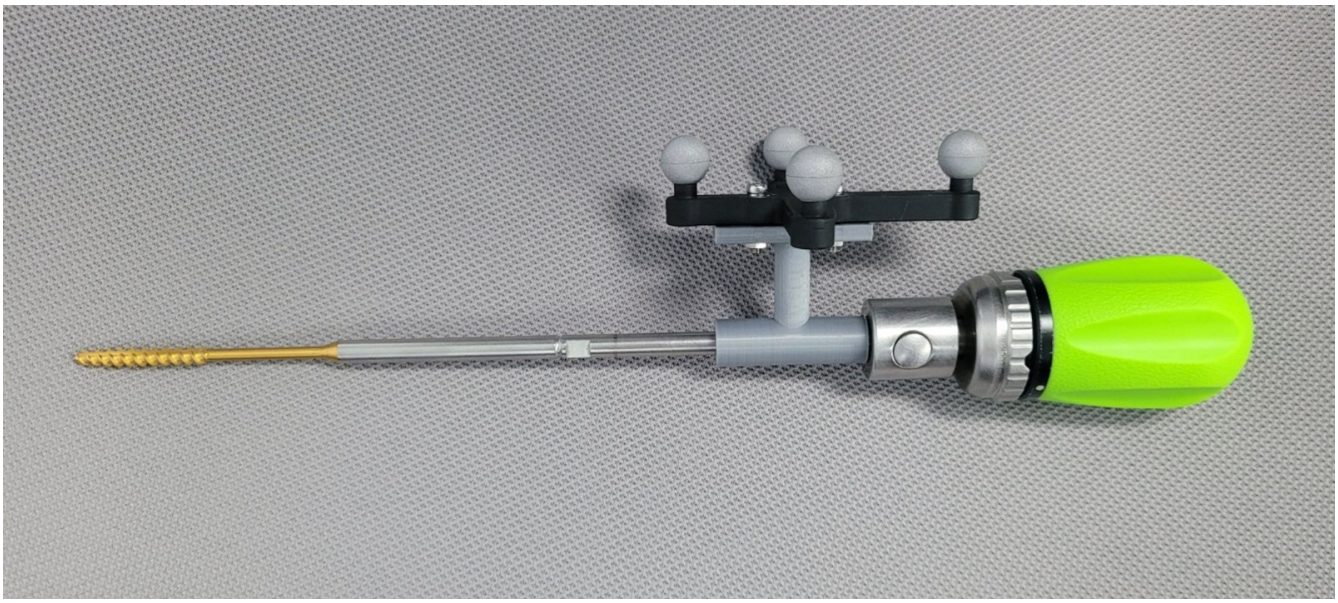


Figure 6. Instrument with a passive-marker adapter used for non-standard and long-instrument calibration.

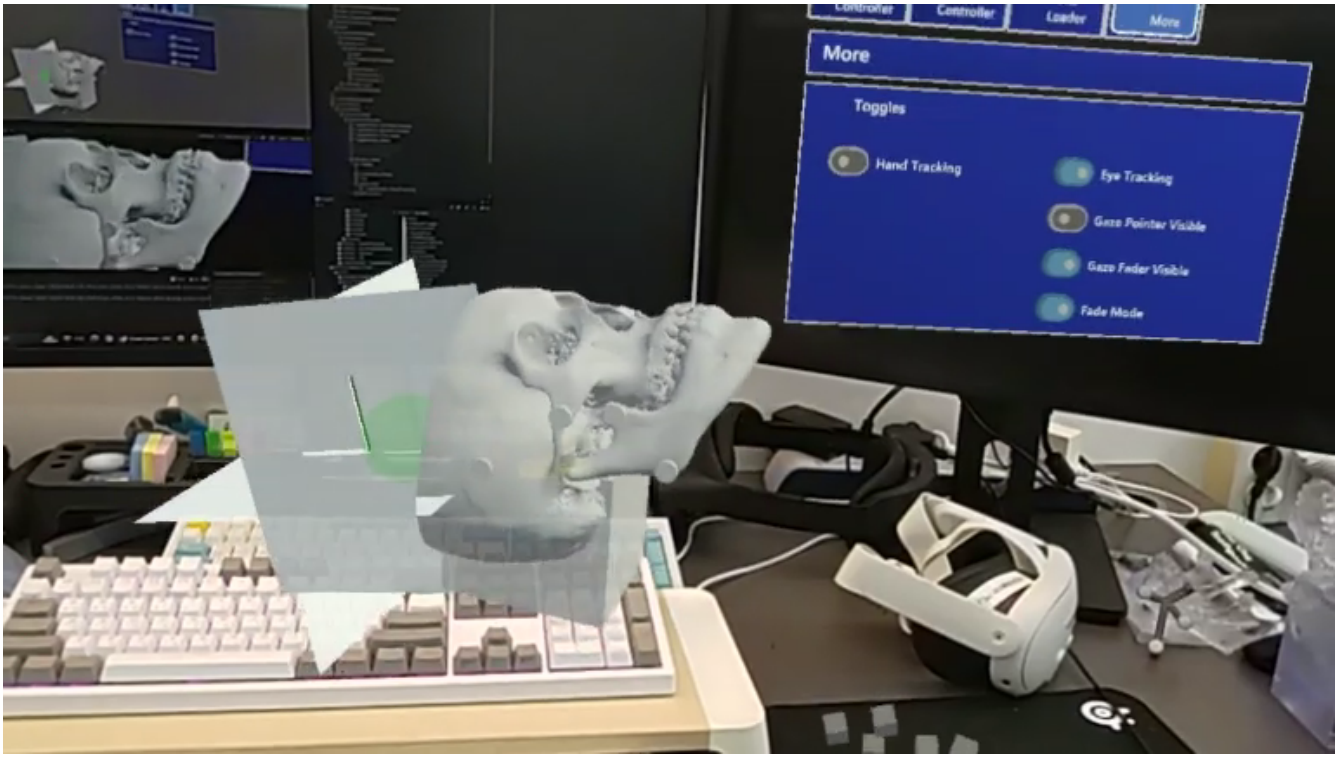


Figure 7. HoloLens spatial presentation with hand- and eye-interaction toggles.

Role, team result, and limits

Team result

The DIGITRACK and Samsung Medical Center research team jointly reviewed the clinical-workflow and requirements context, acceptance criteria, and integration demonstrations.

Limitations

An installer with a checksum manifest exists, but accuracy testing on physical systems and long-duration stability have not been verified. This case does not claim production deployment, use in real surgery, or clinical efficacy, safety, or accuracy.

Collaboration boundary

I worked with researchers on procedure-specific controls and displays, installation feedback, and evaluation preparation. Researchers own the clinical scenarios and research evaluation; I own the assigned software implementation and integration.

Implementation technologies

HoloLens 2 · Optical tracking · 3D Slicer · Unity · MRTK · OpenIGTLink · Holographic Remoting · C++ · Qt 6 · CTest · NSIS · Cython

Primary capability

3D Registration & Computer Vision · Surgical Navigation & Optical Tracking · XR & Spatial Computing

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Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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